

**SIMATS ENGINEERING
ASSESSMENT TOOL 3
CO 4**

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1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.

Star Topology vs Bus Topology:

Feature	Star Topology	Bus Topology
Cost	More expensive due to switch/hub and extra cables	Less expensive because it uses a single backbone cable
Installation	Easy to install and manage	Simple but troubleshooting can be difficult
Reliability	High; failure of one node does not affect others	Low; backbone failure can stop the entire network
Performance	Better performance with fewer collisions	Performance decreases as more devices are added
Scalability	Easy to add new devices	Adding more devices can reduce performance
Maintenance	Easy to identify and fix individual node failures	Fault detection is more difficult
Overall	Best for modern networks	Suitable for small, low-cost networks

Star topology is generally **more reliable, faster, easier to maintain, and scalable**, but it has a higher installation cost. Bus topology is **cheaper and simpler**, but it has lower reliability and performance when many devices are connected.

2. **Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.**

Mesh vs Star Topology:

Feature	Mesh Topology	Star Topology
Fault Tolerance	Very high; multiple alternate paths are available	Moderate; depends on the central device
Link Failure	Data can use another path	Failure of one node affects only that node
Central Device	No central device required	Requires a hub or switch
Reliability	Very high	High, but central device is a single point of failure
Cost	Very expensive due to many connections	Less expensive than mesh
Maintenance	Complex	Easier
Best Use	Critical and high-reliability networks	LANs, offices, schools, and general networks

Mesh topology provides **higher fault tolerance and reliability** because multiple paths are available for communication. Star topology is **simpler, cheaper, and easier to maintain**, but failure of its central hub or switch can affect the entire network. Therefore, **mesh topology is preferred for critical environments where continuous communication is essential**, while star topology is more suitable for normal organizational networks.

3. **Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows,**

making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

Bus vs Mesh Topology:

Feature	Bus Topology	Mesh Topology
Installation	Simple and easy	Complex and time-consuming
Cabling	Uses a single backbone cable	Requires multiple connections between nodes
Cost	Low	Very high
Scalability	Suitable for small networks	Difficult and expensive as nodes increase
Maintenance	Relatively easy	More difficult due to many connections
Complexity	Low	Very high

Bus topology is **simple, inexpensive, and easy to install**, making it suitable for small networks. Mesh topology provides **high reliability and fault tolerance**, but its installation is complex and costly because many connections are required. Therefore, **bus topology is preferred for simple small networks, while mesh topology is preferred when reliability is more important than cost and complexity.**

- Hybrid topology is **highly scalable** because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

Hybrid vs Star Topology

Feature	Hybrid Topology	Star Topology
Scalability	Very high; can combine different topologies and expand flexibly	Moderate; limited by the capacity of the central switch/hub
Expansion	New network segments can be added without major changes	Requires additional hardware when ports are full
Flexibility	High; suitable for different organizational requirements	Limited compared to hybrid
Complexity	More complex to design and manage	Simple to design and manage
Cost	Generally higher	Generally lower
Best Use	Large and growing organizations	Small to medium-sized networks

Hybrid topology provides **greater scalability and flexibility** because different topologies can be combined and new network segments can be added as required. Star

topology is easier and cheaper to manage but its expansion depends on the capacity of the central switch or hub. Therefore, **hybrid topology is more suitable for large and continuously growing networks**, while star topology is suitable for smaller networks.

5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

Maintenance of Network Topologies:

Topology	Maintenance Level	Assessment
Star	Easy	Faults can be quickly identified and isolated to a specific node or cable.
Bus	Difficult	Backbone cable failures can affect the entire network and are difficult to locate.
Mesh	Very Difficult	Many connections make troubleshooting complex and time-consuming.
Hybrid	Moderate	Flexible but requires skilled administration because multiple topologies are combined.

Among these topologies, star topology is the easiest to maintain because faults can be easily isolated. Bus topology is more difficult due to backbone-related failures, while mesh topology is the most complex because of its large number of connections. Hybrid topology has moderate maintenance requirements because it combines different network structures.